

RTL Design

```
always @(posedge i_clk or negedge i_rst_n) begin
    if (!i_rst_n) begin
        o_op_ready_r <= 0;
        o_in_ready_r <= 0;
        o_out_valid_r <= 0;
        o_out_data_r <= 0;

        curr_state <= IDLE;
        curr_op_q <= 0;

        pixel_ctr_q <= 0;

        display_q <= 3'b100;
        image_size_q <= 5'd16;

        output_ctr_q <= 0;

        origin_row_q <= 0;
        origin_col_q <= 0;
    end
end
```

```
376     end else begin
377         // output
378         o_op_ready_r <= o_op_ready_w;
379         o_in_ready_r <= o_in_ready_w;
380         o_out_valid_r <= o_out_valid_w;
381         o_out_data_r <= o_out_data_w;
382
383         curr_state <= next_state;
384         curr_op_q <= curr_op_d;
385
386         display_q <= display_d; // 4, 2, or 1
387
388         // output_ctr_q <= output_ctr_d; // counting output pixels using display size.
389
390         image_size_q <= image_size_d; // display size, 16, 8, or 4
391
392         origin_row_q <= origin_row_d;
393         origin_col_q <= origin_col_d;
394
395         if (i_in_valid && (curr_state == LOAD)) begin
396             pixels[pixel_ctr_q] <= i_in_data;
397             pixel_ctr_q <= pixel_ctr_q + 1'b1;
398         end else if (curr_state != LOAD) begin
399             pixel_ctr_q <= 0;
400         end
401
402         if (curr_state == DONE) begin
403             // if (output_ctr_q <= max_pixels) begin
404                 output_ctr_q <= output_ctr_q + 1;
405             end else begin
406                 output_ctr_q <= 0;
407             end
408         end
409     end
410 end
411 endmodule
```

The above sequential block is responsible for clocked state transition, capturing all registered outputs *_r from their combinational values from *_w. The LOAD state is also done here, when the current state is LOAD and the input instruction is valid, we would take in the input data and store it locally, until we have 256 pixels, the transition from the LOAD state is handled in the top level FSM.

Top Level FSM

```

130     case (curr_state)
131     IDLE : begin
132         o_op_ready_w = 1'b1;
133         //if ((i_op_mode == 4'b0000) && (i_op_valid == 1'b1)) begin
134         // next_state = LOAD;
135         //end
136
137         case ({i_op_valid, i_op_mode})
138         5'b10000 : begin
139             next_state = LOAD;
140         end
141
142         5'b10100, 5'b10101, 5'b10110, 5'b10111 : begin
143             next_state = SHIFT;
144         end
145
146         5'b11000, 5'b11001 : begin
147             next_state = SCALE;
148         end
149
150         5'b11101 : begin
151             next_state = DONE;
152         end
153
154         5'b11110 : begin
155             next_state = DONE;
156         end
157
158         5'b11100 : begin
159             next_state = DONE;
160         end
161
162         default : begin
163             next_state = IDLE;
164         end
165     endcase
166
167     curr_op_d = i_op_mode;
168 end

```

Initially, start off in the IDLE state, based on the operations mode and if the operation is valid, set next_state to the corresponding state.

```

170         LOAD : begin
171             o_op_ready_w = 1'b0;
172             o_in_ready_w = 1'b1;
173             if (pixel_ctr_q == 8'd255 && i_in_valid) begin
174                 next_state = IDLE;
175                 o_in_ready_w = 1'b0;
176             end
177         end
178

```

Based on the current value of pixel_ctr_q, check if 256 pixels are loaded then change the state to IDLE, since no need to display.

```

178
179     SHIFT : begin
180         o_op_ready_w = 1'b0;
181
182         // right
183         if (curr_op_q == 4'b0100) begin
184             if ((origin_col_q + pixel_offset + display_q) < image_size_q) begin
185                 origin_col_d = origin_col_q + pixel_offset;
186             end
187         end
188
189         // left
190         if (curr_op_q == 4'b0101) begin
191             if ((origin_col_q >= pixel_offset)) begin
192                 origin_col_d = origin_col_q - pixel_offset;
193             end
194         end
195
196         // up
197         if (curr_op_q == 4'b0110) begin
198             if ((origin_row_q >= pixel_offset)) begin
199                 origin_row_d = origin_row_q - pixel_offset;
200             end
201         end
202
203         // down
204         if (curr_op_q == 4'b0111) begin
205             if ((origin_row_q + pixel_offset + display_q) < image_size_q) begin
206                 origin_row_d = origin_row_q + pixel_offset;
207             end
208         end
209
210         next_state = DONE;
211     end
212

```

In the SHIFT state, check based on if its LEFT, RIGHT, TOP, or BOTTOM shift. Then check if the new origin will fit within the display size / image_size, that is set based on the scaled version.

```

214         SCALE : begin
215             o_op_ready_w = 1'b0;
216
217             if (curr_op_q == 4'b1000) begin
218                 if (display_q > 3'd1)
219                     display_d = display_q >> 1;
220             end
221
222             if (curr_op_q == 4'b1001) begin
223                 if (display_q < 3'd4)
224                     display_d = display_q << 1;
225             end
226
227             next_state = DONE;
228         end

```

For scale, based on the current image size, the image can be scaled up or down from it, if the display size is bigger than 1 then can be scaled down, if smaller than 4, then scaled up.

```

242     DONE : begin
243         o_op_ready_w = 1'b0;
244         if (output_ctr_q < max_pixels) begin
245             o_out_valid_w = 1'b1;
246
247             case (display_q)
248                 3'd4: o_out_data_w = pixels[{origin_row_q + output_ctr_q[3:2], origin_col_q + output_ctr_q[1:0]}};
249
250                 //3'd2: o_out_data_w = pixels[{origin_row_q + output_ctr_q[1]*2,
251                 //      origin_col_q + output_ctr_q[0]*2}];
252
253                 3'd2: o_out_data_w = pixels[{origin_row_q + row_off_2, origin_col_q + col_off_2}];
254
255                 3'd1: o_out_data_w = pixels[{origin_row_q, origin_col_q}];
256
257                 default: o_out_data_w = 0;
258             endcase
259
260             R = o_out_data_w[23:16];
261             G = o_out_data_w[15:8];
262             B = o_out_data_w[7:0];
263
264             if (curr_op_q == 4'b1101) begin
265                 // fixed point arith. Mult-ed everything by 8 to avoid the fractions. Then div by 8 at the end
266                 Y = ({2'b0,8,1'b0} + {1'b0,6,2'b0} + {3'b0,6} + 11'd4) >> 3;
267
268                 Cb = ({4'b0,8,2'b0} + 12'd1028 - {4'b0,8} - {3'b0,6,1'b0});
269
270                 Cr = ({2'b0,8,2'b0} + 12'd1028 - {2'b0,6,1'b0} - {3'b0,6} - {4'b0,8});
271
272                 o_out_data_w = {Y[7:0], Cb[10:3], Cr[10:3]};
273             end
274
275             if (curr_op_q == 4'b1110) begin
276                 case (display_q)
277                     3'd4: begin
278                         row_temp = {2'b0, origin_row_q + output_ctr_q[3:2]};
279                         col_temp = {2'b0, origin_col_q + output_ctr_q[1:0]};
280

```

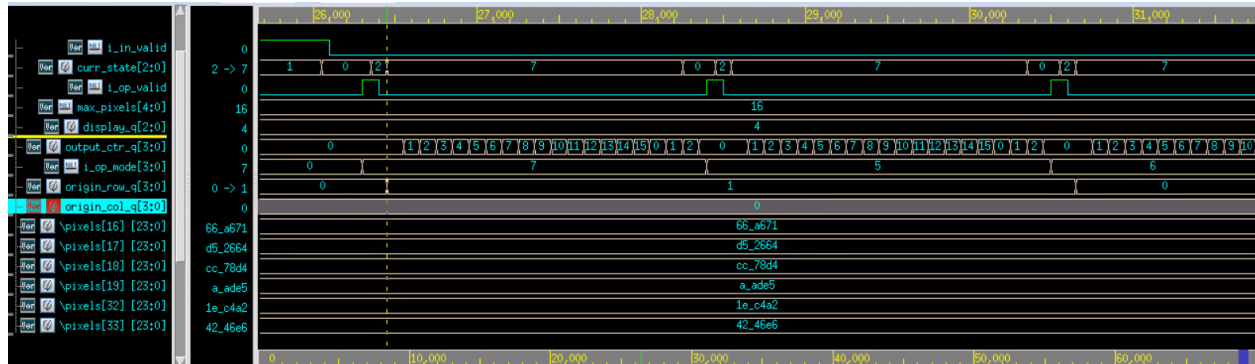
```

280         end
281         3'd2: begin
282             row_temp = {2'b0, origin_row_q + row_off_2};
283             col_temp = {2'b0, origin_col_q + col_off_2};
284         end
285         default: begin
286             row_temp = {2'b0, origin_row_q};
287             col_temp = {2'b0, origin_col_q};
288         end
289     endcase
290
291     shift_temp = {3'b0, pixel_offset};
292
293     p0 = ((row_temp-shift_temp >= 0) && (col_temp-shift_temp >= 0)) ? pixels[(row_temp-shift_temp)*16 + (col_temp-shift_temp)] : 24'd0;
294
295     p1 = ((row_temp-shift_temp >= 0)) ? pixels[(row_temp-shift_temp)*16+col_temp] : 24'd0;
296
297     p2 = ((row_temp-shift_temp >= 0) && (col_temp-shift_temp <= 15)) ? pixels[(row_temp-shift_temp)*16 + (col_temp-shift_temp)] : 24'd0;
298
299     p3 = ((col_temp-shift_temp >= 0)) ? pixels[row_temp*16+(col_temp-shift_temp)] : 24'd0;
300
301     p5 = ((col_temp-shift_temp <= 15)) ? pixels[row_temp*16+(col_temp-shift_temp)] : 24'd0;
302
303     p6 = ((row_temp-shift_temp <= 15) && (col_temp-shift_temp >= 0)) ? pixels[(row_temp-shift_temp)*16 + (col_temp-shift_temp)] : 24'd0;
304
305     p7 = ((row_temp-shift_temp <= 15)) ? pixels[(row_temp-shift_temp)*16+col_temp] : 24'd0;
306
307     p8 = ((row_temp-shift_temp <= 15) && (col_temp-shift_temp <= 15)) ? pixels[(row_temp-shift_temp)*16 + (col_temp-shift_temp)] : 24'd0;
308
309     tempR[7] = (p0[23:16] > 8) ? 1'b1 : 1'b0;
310     tempR[6] = (p1[23:16] > 8) ? 1'b1 : 1'b0;
311     tempR[5] = (p2[23:16] > 8) ? 1'b1 : 1'b0;
312     tempR[4] = (p3[23:16] > 8) ? 1'b1 : 1'b0;
313     tempR[3] = (p5[23:16] > 8) ? 1'b1 : 1'b0;
314     tempR[2] = (p6[23:16] > 8) ? 1'b1 : 1'b0;
315     tempR[1] = (p7[23:16] > 8) ? 1'b1 : 1'b0;
316     tempR[0] = (p8[23:16] > 8) ? 1'b1 : 1'b0;
317
318     tempG[7] = (p0[15:8] > 6) ? 1'b1 : 1'b0;
319     tempG[6] = (p1[15:8] > 6) ? 1'b1 : 1'b0;
320     tempG[5] = (p2[15:8] > 6) ? 1'b1 : 1'b0;
321     tempG[4] = (p3[15:8] > 6) ? 1'b1 : 1'b0;
322     tempG[3] = (p5[15:8] > 6) ? 1'b1 : 1'b0;
323
324
325
326
327     tempB[7] = (p0[7:0] > 8) ? 1'b1 : 1'b0;
328     tempB[6] = (p1[7:0] > 8) ? 1'b1 : 1'b0;
329     tempB[5] = (p2[7:0] > 8) ? 1'b1 : 1'b0;
330     tempB[4] = (p3[7:0] > 8) ? 1'b1 : 1'b0;
331     tempB[3] = (p5[7:0] > 8) ? 1'b1 : 1'b0;
332     tempB[2] = (p6[7:0] > 8) ? 1'b1 : 1'b0;
333     tempB[1] = (p7[7:0] > 8) ? 1'b1 : 1'b0;
334     tempB[0] = (p8[7:0] > 8) ? 1'b1 : 1'b0;
335
336     o_out_data_w = {tempR, tempG, tempB};
337     end
338
339     end else begin
340         o_op_ready_w = 1'b1;
341         o_out_valid_w = 1'b0;
342         next_state = IDLE;
343     end
344     end
345
346     default : begin
347     end
348     endcase
349 end
350

```

The DONE state is the display state for various operations. This sets o_out_valid then sets the output data to be the correct pixel based on the display size from the current origin.

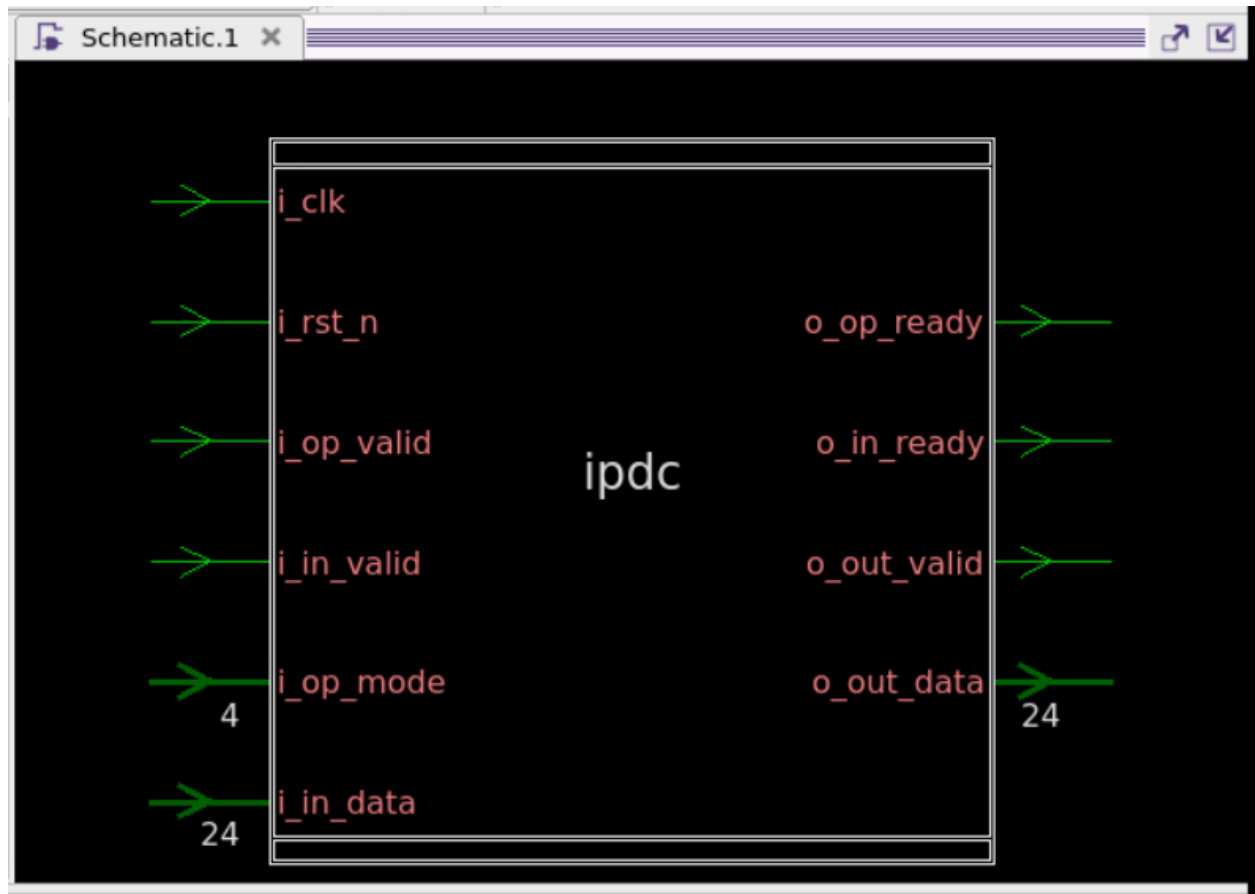
Sample Waveform

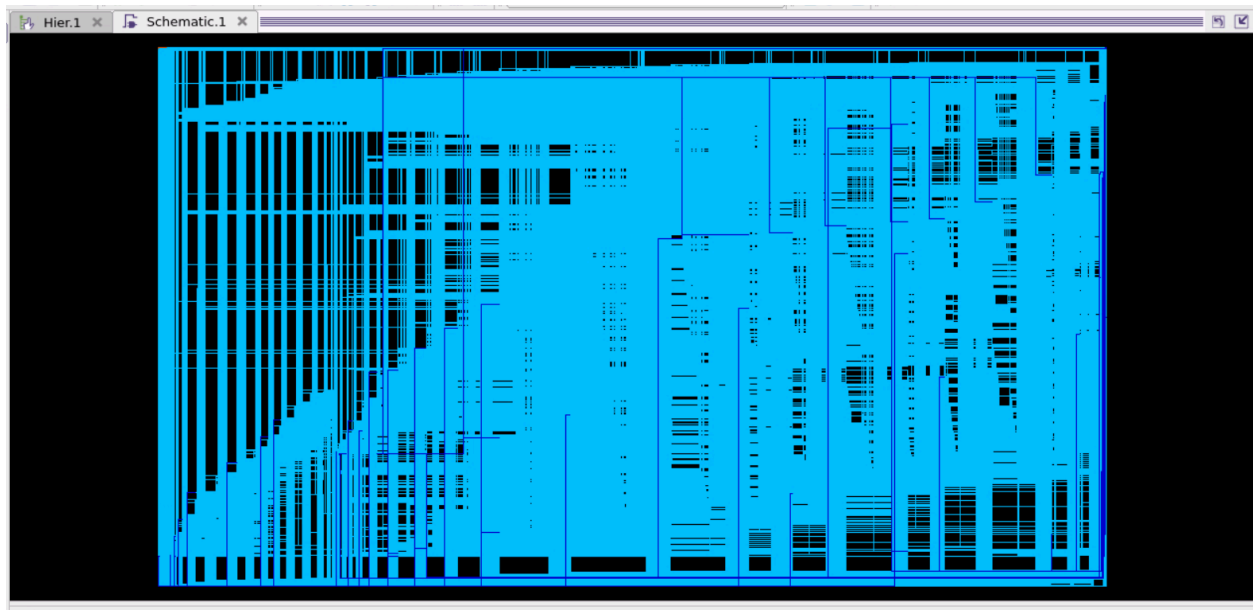


Synthesis

Netlist directory: *Simple-IPDC/MiniProject_03/Lab3_modified/outputs/*

Netlist Filename: *IPDC_syn.v*





Timing

opmode0.dat × indata0.dat × golden0.dat × testbed_temp.v × tb.v × dc_run.tcl × netlist.v × sha256.mmmc × IPDC.timing ×			
U00185/Y (HB1xp67 ASAP7_75t SL)	108.43	1021.86	T
U00177/Y (HB1xp67 ASAP7_75t SL)	115.91	1137.76	f
U34391/Y (OAI22xp33 ASAP7_75t L)	40.38	1178.14	r
U34392/Y (NOR4xp25 ASAP7_75t L)	16.63	1194.77	f
U34403/Y (OR4x1 ASAP7_75t L)	28.14	1222.91	f
U34404/Y (OR5x1 ASAP7_75t L)	33.73	1256.65	f
U34405/Y (AOI22xp33 ASAP7_75t L)	14.20	1270.84	r
U34406/Y (OAI221xp5 ASAP7_75t L)	14.10	1284.95	f
U13008/Y (AND2x2 ASAP7_75t SL)	18.56	1303.51	f
U67100/Y (AO21x1 ASAP7_75t L)	16.02	1319.53	f
U67101/Y (OAI221xp5 ASAP7_75t L)	10.95	1330.49	r
U67109/Y (INVx1 ASAP7_75t SL)	5.52	1336.00	f
U67102/Y (AO221x1 ASAP7_75t L)	23.18	1359.18	f
U67103/Y (OAI221xp5 ASAP7_75t L)	11.50	1370.68	r
U67110/Y (INVx1 ASAP7_75t SL)	5.52	1376.20	f
U67104/Y (AO221x1 ASAP7_75t L)	23.18	1399.37	f
U67105/Y (OAI221xp5 ASAP7_75t L)	11.50	1410.87	r
U67111/Y (INVx1 ASAP7_75t SL)	5.91	1416.78	f
U67107/Y (A201A1xp33 ASAP7_75t L)	11.65	1428.43	r
U67108/Y (OAI21xp33 ASAP7_75t L)	11.26	1439.70	f
U13678/Y (AOI222xp33 ASAP7_75t SL)	16.54	1456.24	r
o_out_data_r_reg[7]/D (DFFASRHQNX1_ASAP7_75t L)	0.00	1456.24	r
data arrival time		1456.24	
clock clk (rise edge)	100000.00	100000.00	
clock network delay (ideal)	0.00	100000.00	
o_out_data_r_reg[7]/CLK (DFFASRHQNX1_ASAP7_75t L)	0.00	100000.00	r
library setup time	-22.41	99977.59	
data required time		99977.59	
data required time		99977.59	
data arrival time		-1456.24	
slack (MET)		98521.35	

The total slack is ≥ 0 . Which means the current design at 1ns clk meets timing requirements.

Critical Path:

Startpoint: display_q_reg[1]
(rising edge-triggered flip-flop clocked by clk)
Endpoint: o_out_data_r_reg[7]
(rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

Area

The screenshot shows a software interface with a file explorer at the top and a summary of area statistics below. The file explorer displays a list of files including 'indata0.dat', 'golden0.dat', 'testbed_temp.v', 'tb.v', 'dc_run.tcl', 'netlist.v', 'sha256.mmmmc', 'IPDC.timing', and 'IPDC.area'. The 'IPDC.area' file is selected. Below the file list, a summary of area statistics is displayed, including the number of ports, nets, cells, combinational cells, sequential cells, macros/black boxes, buf/inv, and references. The total cell area is 6691.870035, and the total area is undefined.

File	Path
asap7sc7p5t_SIMPLE_SLVT_TT_nldm_211120	(File: /home/v72143/mydata/repos/Simple-IPDC/MiniProject_03/Lab3_modified/lib/db/)
asap7sc7p5t_SIMPLE_SLVT_TT_nldm_211120	(File: /home/v72143/mydata/repos/Simple-IPDC/MiniProject_03/Lab3_modified/lib/db/)
asap7sc7p5t_OA_SLVT_TT_nldm_211120	(File: /home/v72143/mydata/repos/Simple-IPDC/MiniProject_03/Lab3_modified/lib/db/)
asap7sc7p5t_OA_SLVT_TT_nldm_211120	(File: /home/v72143/mydata/repos/Simple-IPDC/MiniProject_03/Lab3_modified/lib/db/)
asap7sc7p5t_A0_SLVT_TT_nldm_211120	(File: /home/v72143/mydata/repos/Simple-IPDC/MiniProject_03/Lab3_modified/lib/db/)
asap7sc7p5t_OA_LVT_TT_nldm_211120	(File: /home/v72143/mydata/repos/Simple-IPDC/MiniProject_03/Lab3_modified/lib/db/)
asap7sc7p5t_OA_LVT_TT_nldm_211120	(File: /home/v72143/mydata/repos/Simple-IPDC/MiniProject_03/Lab3_modified/lib/db/)
asap7sc7p5t_A0_LVT_TT_nldm_211120	(File: /home/v72143/mydata/repos/Simple-IPDC/MiniProject_03/Lab3_modified/lib/db/)
asap7sc7p5t_SIMPLE_LVT_TT_nldm_211120	(File: /home/v72143/mydata/repos/Simple-IPDC/MiniProject_03/Lab3_modified/lib/db/)
asap7sc7p5t_SIMPLE_LVT_TT_nldm_211120	(File: /home/v72143/mydata/repos/Simple-IPDC/MiniProject_03/Lab3_modified/lib/db/)
asap7sc7p5t_INVBUF_LVT_TT_nldm_211120	(File: /home/v72143/mydata/repos/Simple-IPDC/MiniProject_03/Lab3_modified/lib/db/)
asap7sc7p5t_INVBUF_LVT_TT_nldm_211120	(File: /home/v72143/mydata/repos/Simple-IPDC/MiniProject_03/Lab3_modified/lib/db/)
asap7sc7p5t_SEQ_LVT_TT_nldm_220123	(File: /home/v72143/mydata/repos/Simple-IPDC/MiniProject_03/Lab3_modified/lib/db/)
asap7sc7p5t_SEQ_SLVT_TT_nldm_220123	(File: /home/v72143/mydata/repos/Simple-IPDC/MiniProject_03/Lab3_modified/lib/db/)
asap7sc7p5t_SEQ_SLVT_TT_nldm_220123	(File: /home/v72143/mydata/repos/Simple-IPDC/MiniProject_03/Lab3_modified/lib/db/)

Statistic	Value
Number of ports:	250
Number of nets:	61001
Number of cells:	60752
Number of combinational cells:	54544
Number of sequential cells:	6202
Number of macros/black boxes:	0
Number of buf/inv:	2078
Number of references:	82
Combinational area:	4878.293063
Buf/Inv area:	111.201661
Noncombinational area:	1813.576972
Macro/Black Box area:	0.000000
Net Interconnect area:	undefined (No wire load specified)
Total cell area:	6691.870035
Total area:	undefined

1

Plain Text ▾ Tab Width: 8 ▾ Ln 1, Col 1 ▾ INS

Power

golden0.dat × testbed_temp.v × tb.v × dc_run.tcl × netlist.v × sha256.mmmc × IPDC.timing × IPDC.area × IPDC.power ×

Global Operating Voltage = 0.7
Power-specific unit information :
Voltage Units = 1V
Capacitance Units = 1.000000ff
Time Units = 1ps
Dynamic Power Units = 1mW (derived from V,C,T units)
Leakage Power Units = 1pW

Attributes

i - Including register clock pin internal power

Cell Internal Power = 86.6524 uW (93%)
Net Switching Power = 6.2444 uW (7%)

Total Dynamic Power = 92.8968 uW (100%)
Cell Leakage Power = 227.8286 uW

Power Group	Internal Power	Switching Power	Leakage Power	Total Power	(%)	Attrs

io_pad	0.0000	0.0000	0.0000	0.0000	(0.00%)	
memory	0.0000	0.0000	0.0000	0.0000	(0.00%)	
black_box	0.0000	0.0000	0.0000	0.0000	(0.00%)	
clock_network	8.0303e-02	0.0000	0.0000	8.0303e-02	(25.04%)	i
register	2.8789e-05	1.0476e-05	1.2752e+08	0.1276	(39.77%)	
sequential	0.0000	0.0000	0.0000	0.0000	(0.00%)	
combinational	6.3185e-03	6.2342e-03	1.0031e+08	0.1129	(35.19%)	

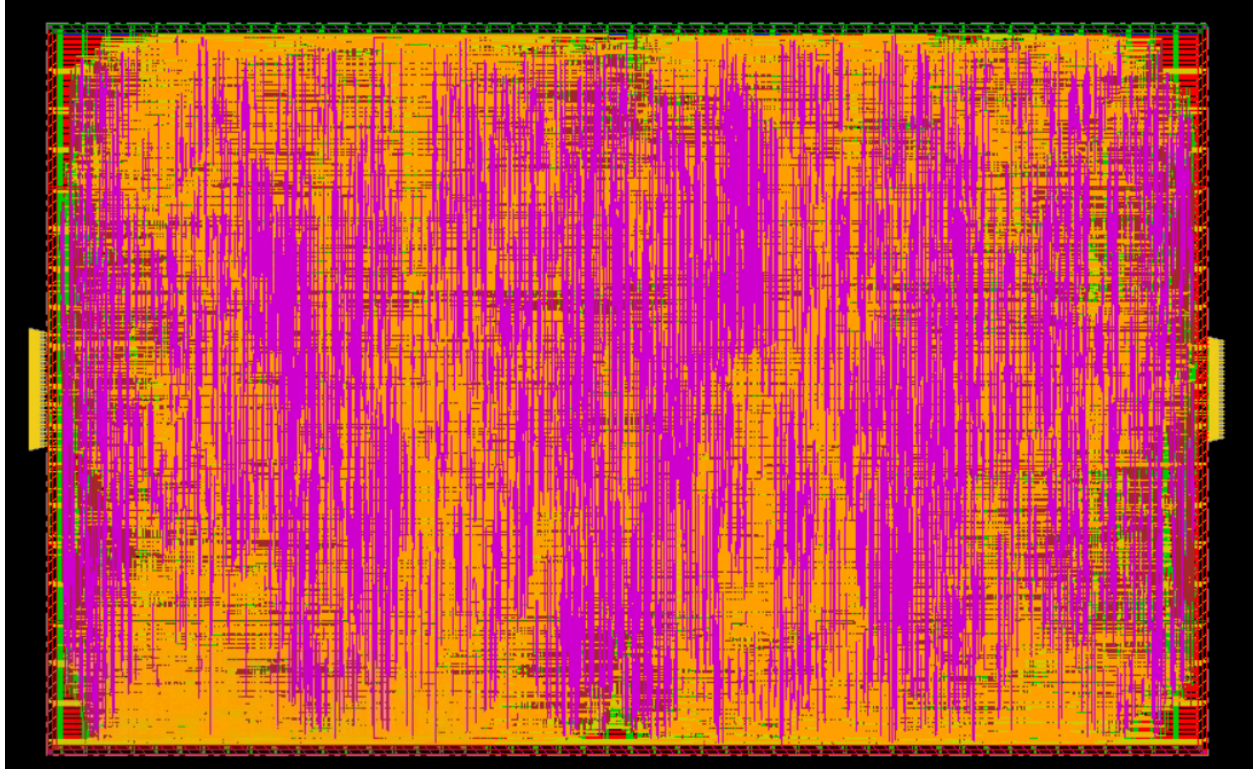
Total	8.6651e-02 mW	6.2446e-03 mW	2.2783e+08 pW	0.3207 mW		

1

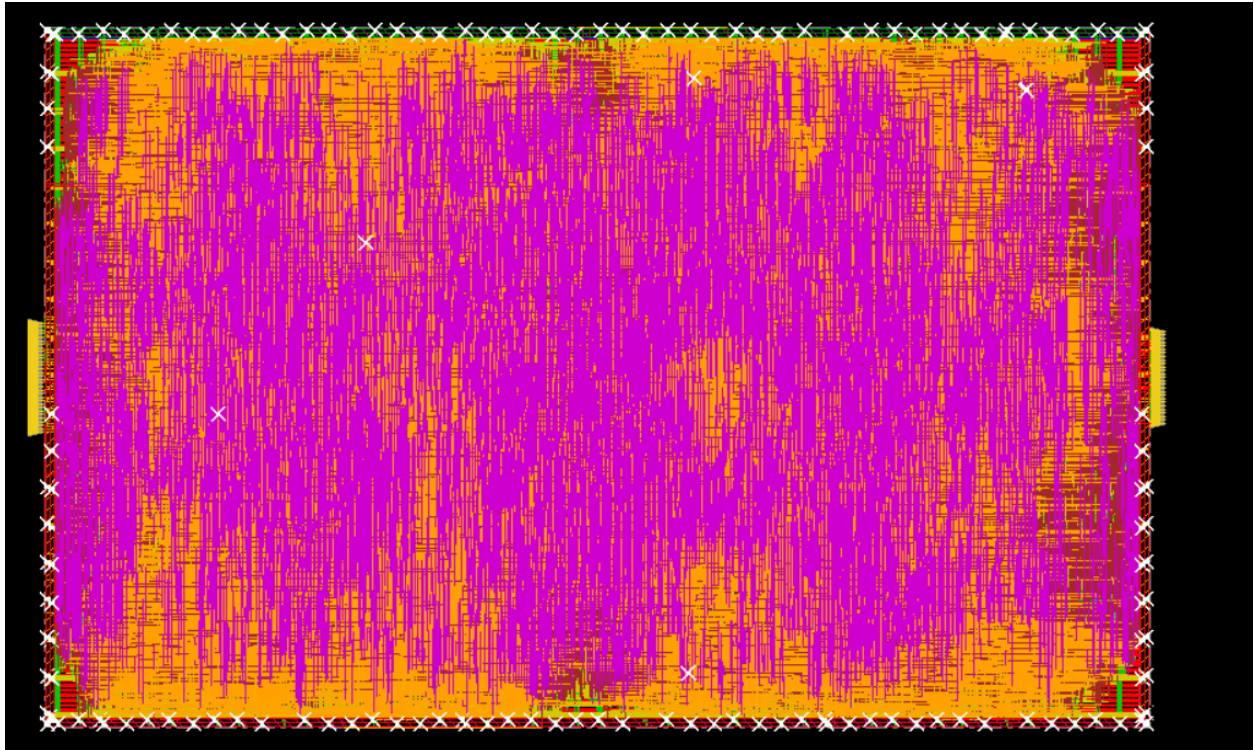
Plain Text ▾ Tab Width: 8 ▾ Ln 17, Col 29 ▾ INS

P&R

Work Directory: Simple-IPDC/MiniProject_03/asap7_reference_design/run_dir
After running place_opt_design and CTS output



After running routeDesign and verify_drc



Scripts

dc_run.tcl for synthesis

```
set search_path
{"/home/v72143/mydata/repos/Simple-IPDC/MiniProject_03/Lab3_modified/lib/db"}

set link_library "asap7sc7p5t_AO_LVT_TT_nldm_211120.db
asap7sc7p5t_AO_SLVT_TT_nldm_211120.db
asap7sc7p5t_INVBUF_LVT_TT_nldm_211120.db
asap7sc7p5t_INVBUF_SLVT_TT_nldm_211120.db
asap7sc7p5t_OA_LVT_TT_nldm_211120.db
asap7sc7p5t_OA_SLVT_TT_nldm_211120.db asap7sc7p5t_SEQ_LVT_TT_nldm_220123.db
asap7sc7p5t_SEQ_SLVT_TT_nldm_220123.db
asap7sc7p5t_SIMPLE_LVT_TT_nldm_211120.db
asap7sc7p5t_SIMPLE_SLVT_TT_nldm_211120.db"

set target_library "asap7sc7p5t_AO_LVT_TT_nldm_211120.db
asap7sc7p5t_AO_SLVT_TT_nldm_211120.db
asap7sc7p5t_INVBUF_LVT_TT_nldm_211120.db
asap7sc7p5t_INVBUF_SLVT_TT_nldm_211120.db
asap7sc7p5t_OA_LVT_TT_nldm_211120.db
asap7sc7p5t_OA_SLVT_TT_nldm_211120.db asap7sc7p5t_SEQ_LVT_TT_nldm_220123.db
asap7sc7p5t_SEQ_SLVT_TT_nldm_220123.db
asap7sc7p5t_SIMPLE_LVT_TT_nldm_211120.db
asap7sc7p5t_SIMPLE_SLVT_TT_nldm_211120.db"

read_file {"../01_RTL/"} -autoread -recursive -format verilog -top ipdc

create_clock -name "clk" -period 100000 -waveform {"0" "50000"} {"i_clk"}

set_fix_hold clk

set_boundary_optimization ""
set_fix_multiple_port_nets -all -buffer_constant
set_max_area 0

check_design

compile -map_effort medium

report_timing -path full -delay max -max_paths 1 -nworst 1 > ./reports/IPDC.timing
report_power > ./reports/IPDC.power
report_area -nosplit > ./reports/IPDC.area

write -hierarchy -format ddc
write_sdc ./outputs/IPDC.sdc
write_sdf -version 1.0 ./outputs/IPDC.sdf
write -format verilog -hierarchy -output ./outputs/IPDC_syn.v
```

PnR Flow:

```
set init_design_uniquify 1
set init_verilog {../run/netlist.v}
set init_design_netlisttype {Verilog}
set init_design_settop {1}
set init_top_cell {ipdc}
set DB_PATH "../db/"
set LEF_PATH "../lef/scaled/"
set TLEF_PATH "../techlef/"
set CELL_LEF "$LEF_PATH/asap7sc7p5t_28_L_4x_220121a.lef
$LEF_PATH/asap7sc7p5t_28_SL_4x_220121a.lef
$LEF_PATH/asap7sc7p5t_28_R_4x_220121a.lef"
set TECH_LEF $TLEF_PATH/asap7_tech_4x_201209.lef
set init_lef_file "$TECH_LEF $CELL_LEF"
set fp_core_cntl {aspect}
set fp_aspect_ratio {1.0000}
set extract_shrink_factor {1.0}
set init_assign_buffer {0}
set init_pwr_net {VDD}
set init_gnd_net {VSS}
set init_cpf_file {}
set init_mmmc_file {../scripts/sha256.mmmc}
init_design

setDesignMode -process 7 -node N7
setMultiCpuUsage -localCpu 4
setDesignMode -bottomRoutingLayer 2
setDesignMode -topRoutingLayer 7
set FP_RING_OFFSET 0.384
set FP_RING_WIDTH 2.176
set FP_RING_SPACE 0.384
set FP_RING_SIZE [expr {$FP_RING_SPACE + 2*$FP_RING_WIDTH +
$FP_RING_OFFSET + 1.1}]
set FP_TARGET 360
set FP_MUL 8
set cellheight [expr 0.270 * 4]
set cellhgrid 0.216
set fpxdim [expr $cellhgrid * $FP_TARGET * $FP_MUL]
set fpydim [expr $cellheight * $FP_TARGET]
fpGetSnapRule
floorPlan -site asap7sc7p5t -s $fpxdim $fpydim $FP_RING_SIZE $FP_RING_SIZE
$FP_RING_SIZE $FP_RING_SIZE -noSnap

add_tracks -snap_m1_track_to_cell_pins
add_tracks -mode replace -offsets {M5 vertical 0}
deleteAllFPObjets
addWellTap -cell TAPCELL_ASAP7_75t_L -cellInterval 12.960 -inRowOffset 1.296
```

```

setPinAssignMode -pinEditInBatch true
editPin -fixOverlap 1 -unit MICRON -spreadDirection clockwise -side Left -layer 3 -spreadType
center -spacing 2.016 -pin {i_clk i_rst_n i_op_valid i_in_valid {i_op_mode[0]} {i_op_mode[1]}
{i_op_mode[2]} {i_op_mode[3]} {i_in_data[0]} {i_in_data[1]} {i_in_data[2]} {i_in_data[3]}
{i_in_data[4]} {i_in_data[5]} {i_in_data[6]} {i_in_data[7]} {i_in_data[8]} {i_in_data[9]}
{i_in_data[10]} {i_in_data[11]} {i_in_data[12]} {i_in_data[13]} {i_in_data[14]} {i_in_data[15]}
{i_in_data[16]} {i_in_data[17]} {i_in_data[18]} {i_in_data[19]} {i_in_data[20]} {i_in_data[21]}
{i_in_data[22]} {i_in_data[23]}}
editPin -fixOverlap 1 -unit MICRON -spreadDirection clockwise -side Right -layer 3
-spreadType center -spacing 2.016 -pin {o_op_ready o_in_ready o_out_valid {o_out_data[0]}
{o_out_data[1]} {o_out_data[2]} {o_out_data[3]} {o_out_data[4]} {o_out_data[5]}
{o_out_data[6]} {o_out_data[7]} {o_out_data[8]} {o_out_data[9]} {o_out_data[10]}
{o_out_data[11]} {o_out_data[12]} {o_out_data[13]} {o_out_data[14]} {o_out_data[15]}
{o_out_data[16]} {o_out_data[17]} {o_out_data[18]} {o_out_data[19]} {o_out_data[20]}
{o_out_data[21]} {o_out_data[22]} {o_out_data[23]}}
editPin -snap TRACK -pin *
setPinAssignMode -pinEditInBatch false
legalizePin
globalNetConnect VDD -type pgpin -pin VDD -inst *
globalNetConnect VSS -type pgpin -pin VSS -inst *

setAddRingMode -ring_target default -extend_over_row 0 -ignore_rows 0 -avoid_short 0
-skip_crossing_trunks none -stacked_via_top_layer Pad -stacked_via_bottom_layer M1
-via_using_exact_crossover_size 1 -orthogonal_only true -skip_via_on_pin {standardcell}
-skip_via_on_wire_shape {noshape}
addRing -nets {VDD VSS} -type core_rings -follow core -layer {top M7 bottom M7 left M6 right
M6} -width $FP_RING_WIDTH -spacing $FP_RING_SPACE -offset $FP_RING_OFFSET
-center 0 -threshold 0 -jog_distance 0 -snap_wire_center_to_grid None

addStripe -skip_via_on_wire_shape blockring -direction horizontal -set_to_set_distance [expr
2*$cellheight] -skip_via_on_pin Standardcell -stacked_via_top_layer M1 -layer M2 -width
0.072 -nets {VDD} -stacked_via_bottom_layer M1 -start_from bottom
-snap_wire_center_to_grid Grid -start_offset 0 -stop_offset 0

addStripe -skip_via_on_wire_shape blockring -direction horizontal -set_to_set_distance [expr
2*$cellheight] -skip_via_on_pin Standardcell -stacked_via_top_layer M1 -layer M2 -width
0.072 -nets {VSS} -stacked_via_bottom_layer M1 -start_from bottom
-snap_wire_center_to_grid Grid -start_offset [expr $cellheight] -stop_offset 0

set m3pwrwidth 0.936
set m3pwrspacing 0.360
set m3pwrset2setdist 12.960
addStripe -skip_via_on_wire_shape Noshape -set_to_set_distance $m3pwrset2setdist
-skip_via_on_pin Standardcell -stacked_via_top_layer Pad -spacing $m3pwrspacing
-xleft_offset 0.360 -layer M3 -width $m3pwrwidth -nets {VDD VSS} -stacked_via_bottom_layer
M2 -start_from left
set m4pwrwidth 0.864
set m4pwrspacing 0.864
set m4pwrset2setdist 21.6

```

```
addStripe -skip_via_on_wire_shape Noshape -direction horizontal -set_to_set_distance  
$m4pwrset2setdist -skip_via_on_pin Standardcell -stacked_via_top_layer M7 -spacing  
$m4pwrspacing -layer M4 -width $m4pwrwidth -nets {VDD VSS} -stacked_via_bottom_layer  
M3 -start_from bottom
```

```
setSroutemode -reset  
setSroutemode -viaConnectToShape { noshape }  
sroutemode -connect { corePin } -layerChangeRange { M1(1) M7(1) } -blockPinTarget {  
nearestTarget } -floatingStripeTarget { blockring padring ring stripe ringpin blockpin followpin }  
-deleteExistingRoutes -allowJogging 0 -crossoverViaLayerRange { M1(1) Pad(10) } -nets {  
VDD VSS } -allowLayerChange 0 -targetViaLayerRange { M1(1) Pad(10) }  
editPowerVia -add_vias 1 -orthogonal_only 0
```

```
colorizePowerMesh  
verify_drc
```